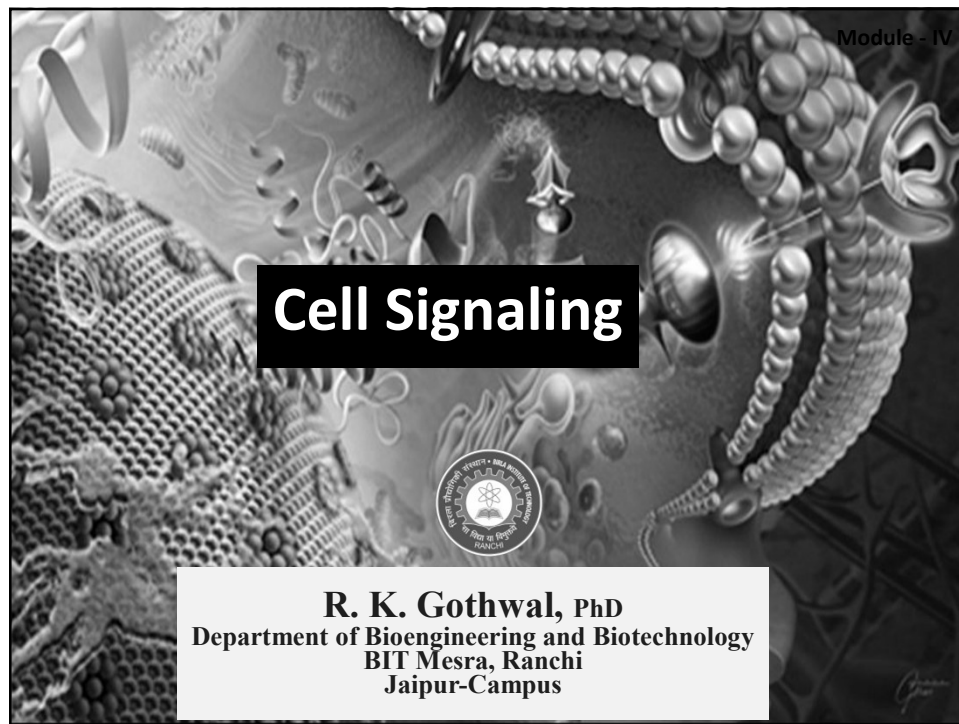
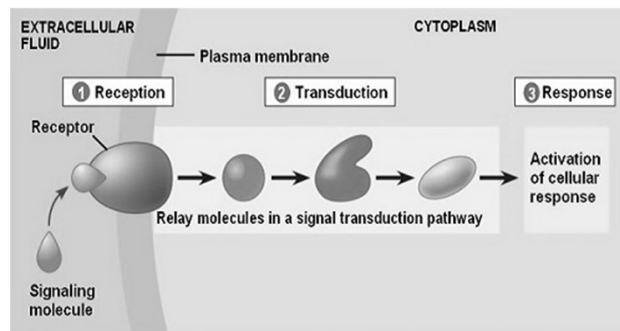




Cell Signaling

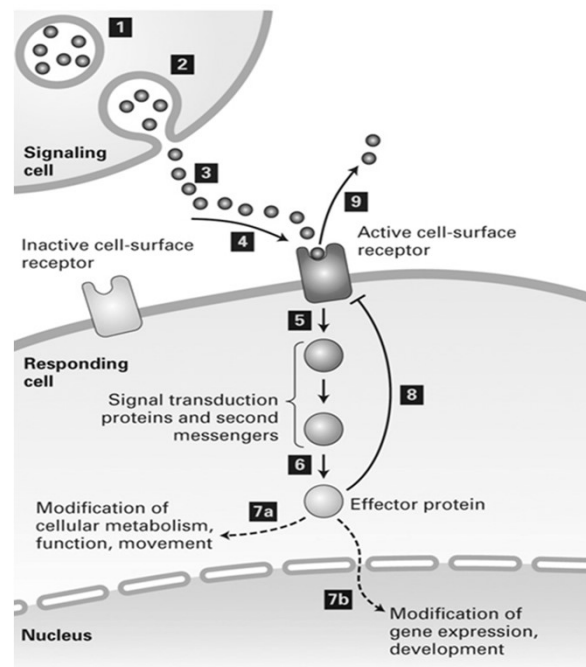
- Cell signaling refers to the cell-cell communications. That is, the way by which the cells communicate with each other.
- Response to stimulus is the nature of Living beings. All live organisms keep receiving signals (external cues) from external environments or surroundings through their sensory organs. These signals transmit to the brain to take further actions.
- The cellular response to these stimulus is carried out by a **series of biological events** which involve different cells to communicate and generate response according to the commands received from brain.



Cell Signaling

- In simple words one cell (**signaling cell**) sends signals (or information) to another cell (**responding cell**) to take action or generate response.
- Mostly the signal or information is transferred in the form of **chemical message** (also called **ligand molecule**). This message is received by a specific group of cells having **receptors** on their surface.
- Receptors on surface receive the signals and starts the **signal transduction** pathway which as a results alters the gene expression levels of responding proteins.

Cell Signaling and Signal Transduction

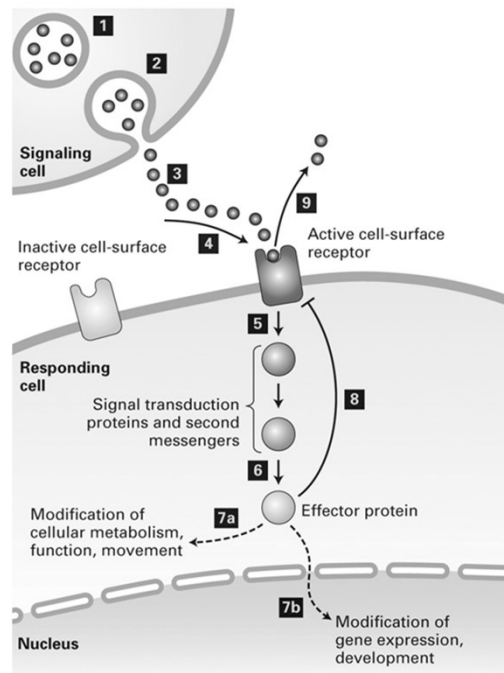


Signal Transduction Pathway

- Signal transduction refers to the overall process of converting extracellular signals into intracellular responses.
- A **signal transduction pathway** represents the flow of information in a cell. The pathway starts when a signaling molecule or **ligand** binds to its **receptor** protein.
- The receptor-ligand interaction triggers a change in the receptor that is carried or **transduced** to the next protein in the pathway.
- **Receptors and signal transduction systems have evolved to detect and respond to hormones, growth factors, neurotransmitters, pheromones, oxygen, nutrients, light, touch, heat, etc.**
- There are an enormous number of signal molecules and receptors in cells. In contrast, there are relatively few types of intracellular signal transduction systems.

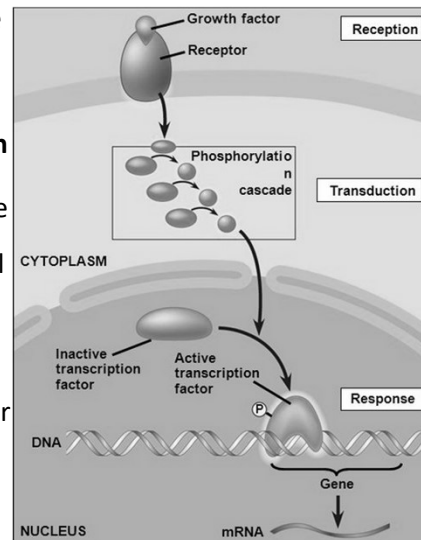
Signal Transduction

- Key players in signal transduction are **signaling molecules, receptors, signal transduction proteins** and **second messengers**, and **effector proteins**.
- Cells respond to signals by changing the activity of existing enzymes (fast) and/or the levels of expression of enzymes and cell components (slower) by gene regulation (Steps 7a & 7b).



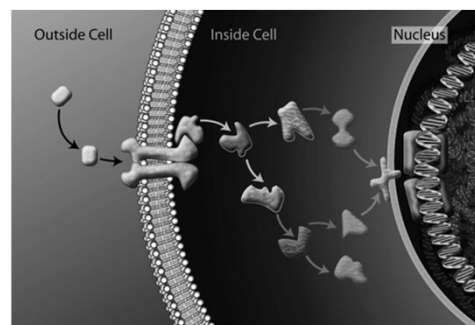
Signal Transduction Pathway

- **Example:** Protein Tyrosine kinase pathway
 - In this, activation of the kinase function of the receptor by ligand binding results in **phosphorylation** of an **adaptor** or other protein.
 - If the next protein in the sequence is an inactive kinase, then the addition of a phosphate group will alter its conformation, converting it into an active kinase.
 - The resulting **phosphorylation cascade** will eventually result in a change in the cell's behavior either by modifying the activity of proteins already present within the cytoplasm or by altering gene expression.



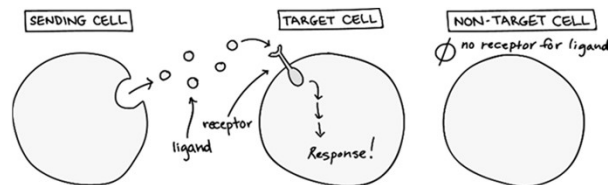
Cell Signaling Facts

- All cells have the capacity to achieve this to some degree, albeit with a wide variation in purpose, mechanism, and response.
- Cells typically communicate using **chemical signals**.
- Signal transduction pathway converts the signals into the action or cellular response which mostly alters the gene expression.
- Higher degree of similarity in signaling mechanism was observed among a range of species in eukaryotes.



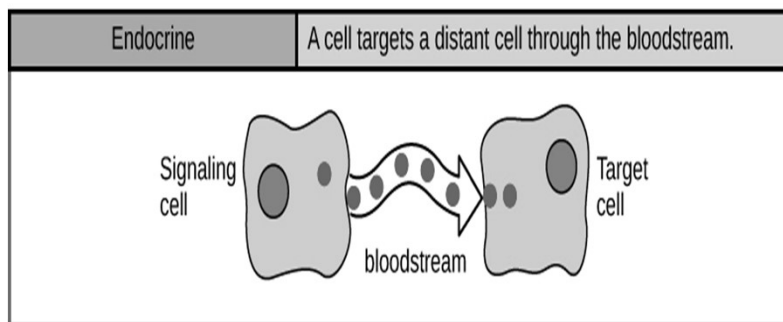
Types of Cell Signaling

- There are four basic categories of chemical signaling found in multicellular organisms:
 - Paracrine signaling
 - Autocrine signaling
 - Endocrine signaling
 - Cell-to-cell direct contact signaling
- The main difference between the different categories of signaling is the **distance** that the signal travels through the organism to reach the target cell.



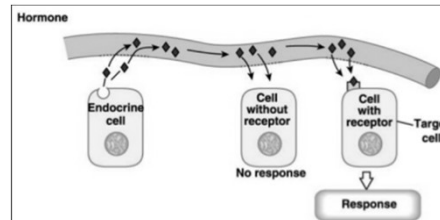
Endocrine Signaling

- Endocrine signaling acts over long distances within the organism (e.g., insulin by endocrine gland).
- A cell targets a distant cell through the bloodstream.
- A signaling molecule is released by one cell, then travels through the bloodstream to bind to receptors on a distant target cell elsewhere in the body.



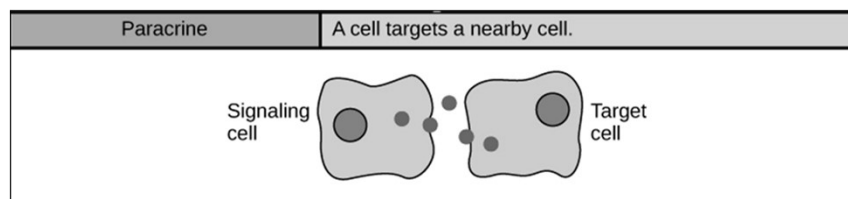
Endocrine Signaling - Example

- In long-distance **endocrine signaling**, signals are produced by specialized cells and released into the bloodstream, which carries them to target cells in distant parts of the body. Signals that are produced in one part of the body and travel through the circulation to reach far-away targets are known as **hormones**.
- In humans, endocrine glands that release hormones include the thyroid, the hypothalamus, and the pituitary, as well as the gonads (testes and ovaries) and the pancreas. Each endocrine gland releases one or more types of hormones, many of which are master regulators of development and physiology.
- For example, the pituitary releases **growth hormone (GH)**, which promotes growth, particularly of the skeleton and cartilage. Like most hormones, GH affects many different types of cells throughout the body. However, cartilage cells provide one example of how GH functions: it binds to receptors on the surface of these cells and encourages them to divide.



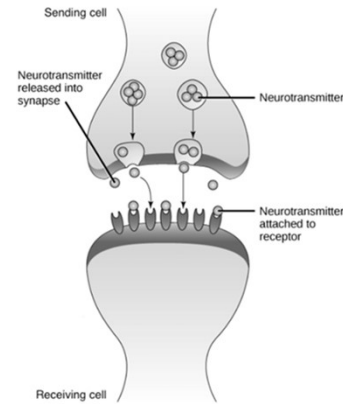
Paracrine Signaling

- Paracrine signaling acts over very short distances, for example between neighboring cells.
 - Example: Neurotransmitters and developmental signals typically act in this manner.
- Paracrine signaling allows cells to locally coordinate activities with their neighbours. Paracrine signals are especially important during development, when they allow one group of cells to tell a neighbouring group of cells what cellular identity to take on. Example: spinal cord development
- The conduction of an electric impulse from one nerve cell to another or from one nerve cell to an effector or muscle cell occurs via paracrine way.



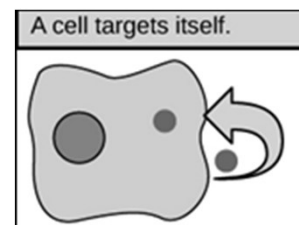
Paracrine Signaling – Synaptic signaling

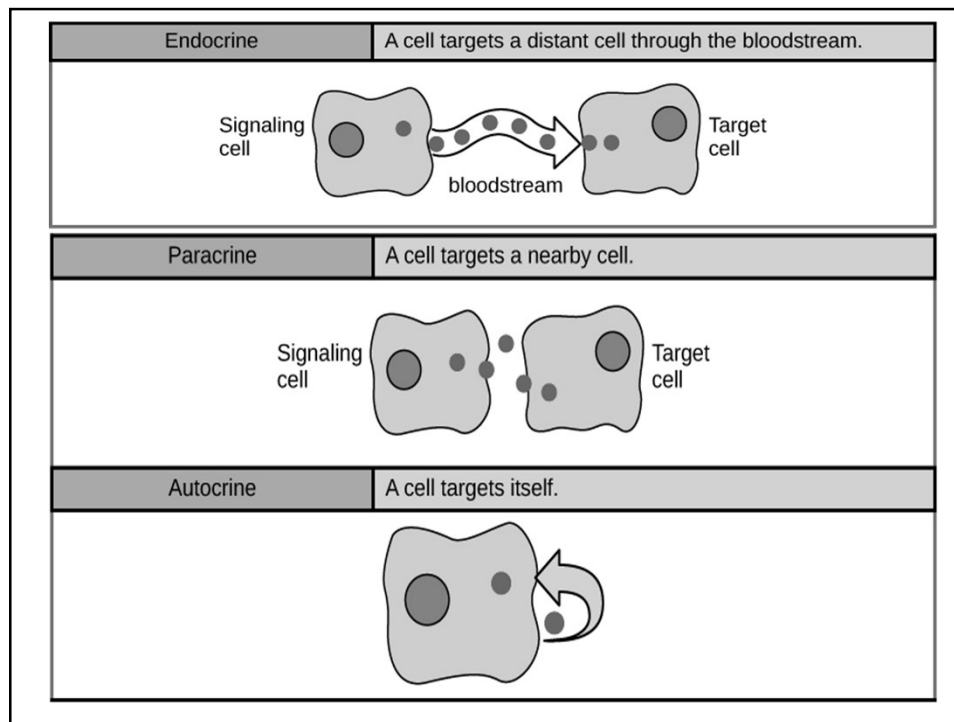
- In this type of signaling nerve cells transmit signals. This process is named for the **synapse**, the junction between two nerve cells where signal transmission occurs.
- When the sending neuron fires, an electrical impulse moves rapidly through the cell, traveling down a long, fiber-like extension called an axon.
- When the impulse reaches the synapse, it triggers the release of ligands called **neurotransmitters**, which quickly cross the small gap between the nerve cells.
- When the neurotransmitters arrive at the receiving cell, they bind to receptors and cause a chemical change inside of the cell (often, opening ion channels and changing the electrical potential across the membrane).
- The neurotransmitters that are released into the chemical synapse are quickly degraded or taken back up by the sending cell. This "resets" the system so they synapse is prepared to respond quickly to the next signal.



Autocrine Signaling

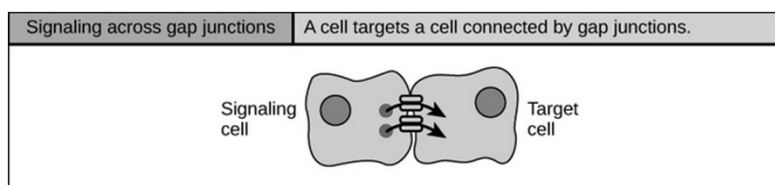
- In **autocrine signaling**, a cell signals to itself, releasing a ligand that binds to receptors on its own surface (or, depending on the type of signal, to receptors inside of the cell).
- Many growth factors act in this fashion, and cultured cells often secrete growth factors that stimulate their growth and proliferation.
- This may seem like an odd thing for a cell to do, but autocrine signaling plays an important role in many processes.
 - For instance, autocrine signaling is important during development, helping cells take on and reinforce their correct identities.
 - From a medical standpoint, autocrine signaling is important in cancer and is thought to play a key role in metastasis (the spread of cancer from its original site to other parts).
- In many cases, a signal may have both autocrine and paracrine effects, binding to the sending cell as well as other similar cells in the area.





Cell-to-Cell Direct Contact Signaling

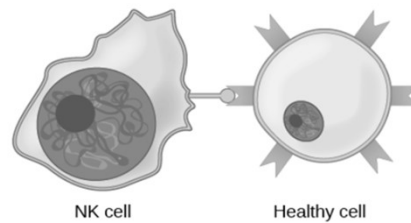
- In this type of signaling systems, cells communicate directly via cell-to-cell contact through cell membrane.
- It involves plasma membrane-attached proteins for communication.
- **Gap junctions** in animal cells and **plasmodesmata** in plant cells are example of such cellular communication where a tiny channel directly connect neighbouring cells and transduce signal through **diffusion**.
- The transfer of signaling molecules transmits the current state of one cell to its neighbour. This allows a group of cells to coordinate their response to a signal that only one of them may have received.
- Small molecules, such as calcium ions (Ca^{2+}), are able to move between cells, but large molecules like proteins and DNA cannot fit through the channels without special assistance.



Cell-to-Cell Direct Contact Signaling

- In another form of direct signaling, two cells may bind to one another because they carry complementary proteins on their surfaces.
- When the proteins bind to one another, this interaction changes the shape of one or both proteins, transmitting a signal.
- This kind of signaling is especially important in the immune system, where immune cells use cell-surface markers to recognize “self” cells (the body's own cells) and cells infected by pathogens

A natural killer (NK) immune cell recognizes a healthy cell of the body by binding to a "self" marker on the cell's surface.



Types of Cell Signaling

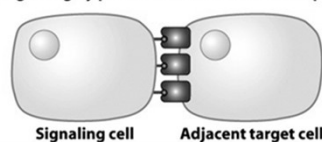
(c) Autocrine signaling



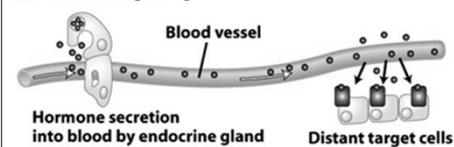
Key:

- Extracellular signal
- Receptor
- ↑ Membrane-attached signal

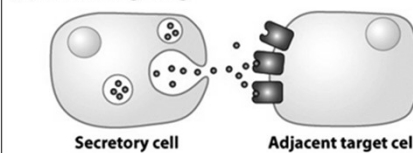
(d) Signaling by plasma membrane-attached proteins



(a) Endocrine signaling



(b) Paracrine signaling

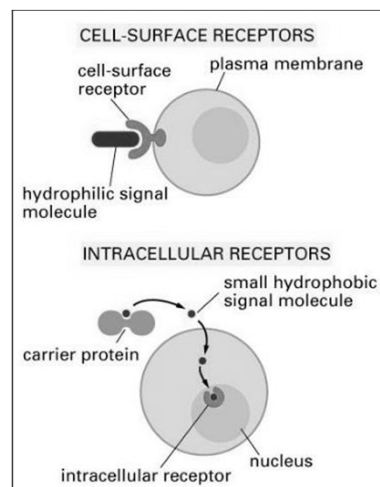


Signaling Receptors and Ligands

- **Receptor** is a molecule which receives specific chemical signal in cellular communication.
- These receptors either found on the surface of a cell (**cell-surface receptors**) or inside the cell in cytoplasm (**Intracellular or nuclear receptors**).
- A molecule which binds to a receptor is called a **ligand**. Ligand may be a peptide (short protein) or other small molecules such as neurotransmitter, a hormone, a drug or a toxin.
- The accurate binding of ligand to receptors opens or elicit a biochemical pathway as a response.

Types of Receptors

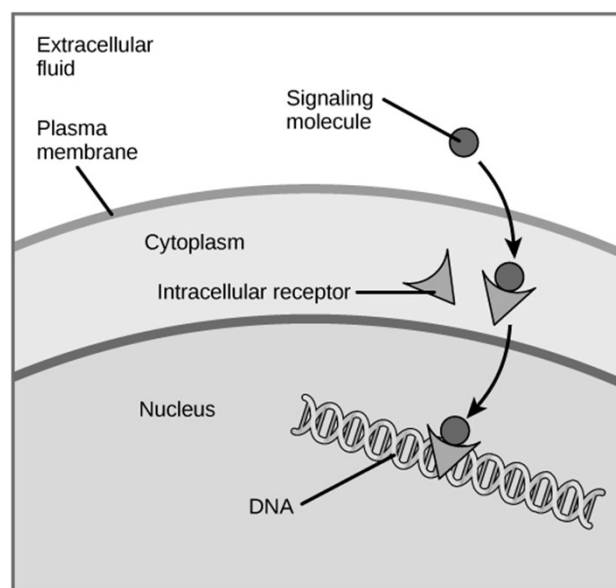
- There are two classes of receptors
 - **Intracellular**
 - Found inside the cell
 - **Extracellular**
 - Found on cell surface



Intracellular Receptors

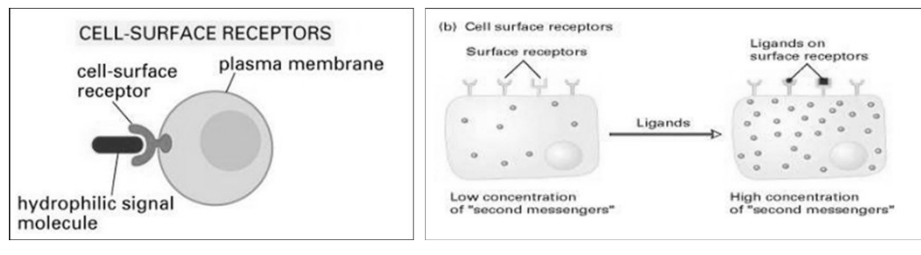
- These are mostly proteins and found inside the cell, usually in cytoplasm or nucleus.
- The signal (or ligand) is very small and hydrophobic (water-hating) in nature.
- Small size (of ligand) enable it to cross the plasma membrane and reach their receptors
- Example
 - Sex hormones - estrogen and testosterone
- When a hormone enters a cell and binds to its receptor, it causes the receptor to change shape, allowing the receptor-hormone complex to enter the nucleus (if it wasn't there already) and regulate gene activity.
- Binding of ligand to receptors makes a new complex. This receptor-ligand complex then bind to specific location on DNA from where it controls the level of its transcription.

Intracellular Receptors



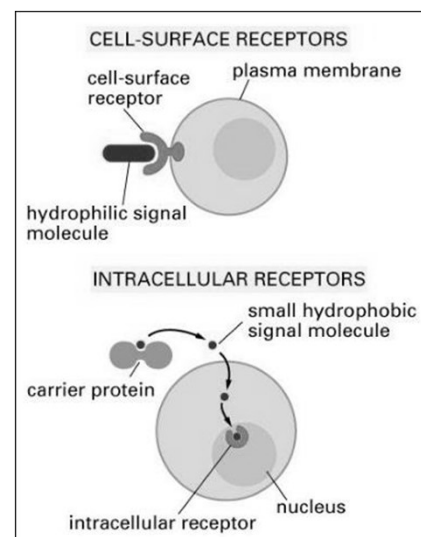
Extracellular Receptors

- Also called **Cell-Surface Receptors**.
- They are membrane-anchored proteins that bind to ligands on the outside surface of the cell.
- In this type of signaling, the ligand does not need to cross the plasma membrane. Many different kinds of molecules (including large, hydrophilic or "water-loving" ones) may act as ligand.
- Most common extracellular signaling molecules are hormones, neurotransmitters, cytokines, growth factors and cell recognition molecules.
- These molecules attached to the receptor and triggers the signal transduction pathway.



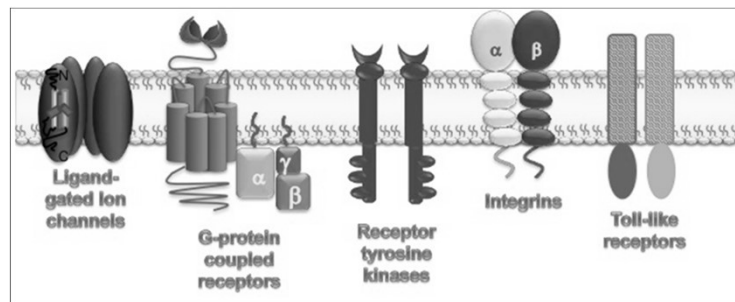
Intra- and Extra-cellular receptors

- Cell surface receptors
 - Receives hydrophilic Signals
 - Membrane bound
 - Signals released into extracellular space – local (paracrine) or long-distance (endocrine)
- Intracellular receptors
 - Receives hydrophobic signals on carrier proteins
 - Pass directly through cell membrane
 - Receptors present in cytosol or nucleus



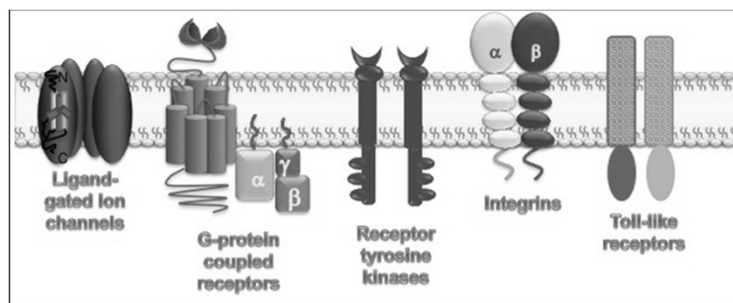
Cell-surface receptors

- A typical cell-surface receptor has three different **domains**, or protein regions:
 - an extracellular ("outside of cell") ligand-binding domain,
 - a hydrophobic domain extending through the membrane, and
 - an intracellular ("inside of cell") domain, which often transmits a signal.



Cell-surface receptors

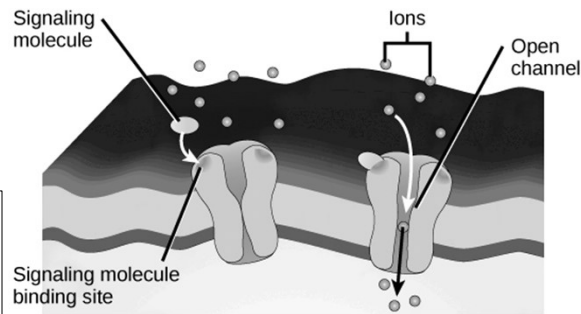
- The size and structure of these regions can vary a lot depending on the type of receptor, and the hydrophobic region may consist of multiple stretches of amino acids that criss-cross the membrane.
- There are many kinds of cell-surface receptors, but here we'll consider only at three common types:
 1. **Ligand-gated ion channels,**
 2. **G protein-coupled receptors, and**
 3. **Receptor tyrosine kinases**



1. Ligand-Gated Ion Channels

- **Ligand-gated ion channels** are ion channels that can open in response to the binding of a ligand.
- To form a channel, this type of cell-surface receptor has a membrane-spanning region with a hydrophilic (water-loving) channel through the middle of it.
- The channel lets ions to cross the membrane without having to touch the hydrophobic core of the phospholipid bilayer.
- Changes in ion levels inside the cell can change the activity of other molecules, such as ion-binding enzymes and voltage-sensitive channels, to produce a response.
- **Examples** - Neurons, or nerve cells, have ligand-gated channels that are bound by neurotransmitters.

When a ligand binds to the extracellular region of the channel, the protein's structure changes in such a way that ions of a particular type, such as Ca^{2+} or Cl^- , can pass through.

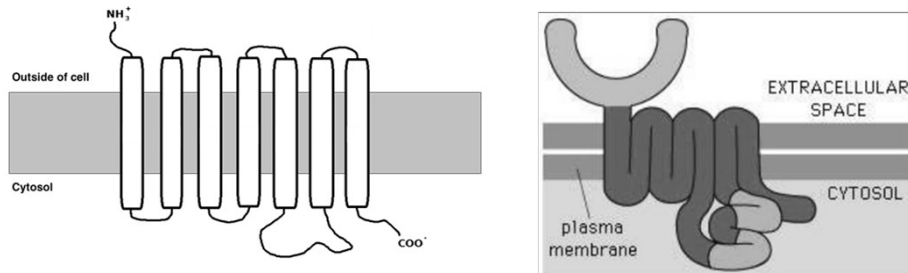


2. G protein-Coupled Receptors (GPCRs)

- **G protein-coupled receptors (GPCRs)** are integral membrane proteins that possess seven membrane-spanning domains (or transmembrane helices).
- These receptors transmit signals (ligand) inside the cell through a type of protein called a **G protein**.
- G Protein is a trimeric protein having three subunits, namely **alpha (α)**, **beta (β)** and **gamma(γ) subunits**.
- These receptors activate a **G protein-Ligand** binding.
 - ✓ The alpha subunit can bind with guanosine diphosphate (GDP), which causes phosphorylation of GDP to guanosine triphosphate (GTP). Phosphorylation activates the alpha (α) subunit to dissociate from beta (β) and gamma(γ) subunits.
 - ✓ The activated alpha (α) subunit can further affect intracellular signaling proteins and target functional protein directly.
- GPCRs are a large family of cell surface receptors that share a common structure and method of signaling.

G protein-coupled receptors (GPCRs)

Cell Surface Receptor



G protein-coupled receptor (GPCR)

GPCRs have seven membrane-spanning domains, as shown by the seven segments crossing the grey region that represents the plasma membrane.

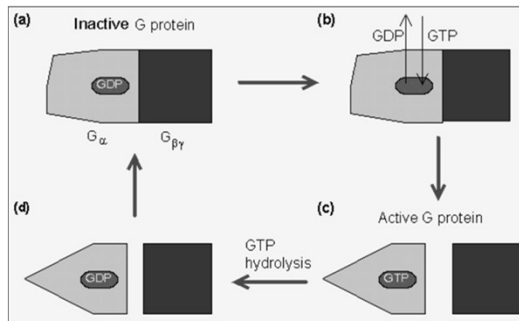
G protein-coupled receptors

- GPCRs are diverse and bind many different types of ligands.
- One particularly interesting class of GPCRs is the **odorant** (scent) receptors.
 - There are about 800 of them in humans, and each binds its own “scent molecule” – such as a particular chemical in perfume, or a certain compound released by rotting fish – and causes a signal to be sent to the brain, making us smell.



G protein – *Active or inactive state*

- G proteins come in different types, but they all bind the nucleotide guanosine triphosphate (GTP), which they can break down (hydrolyze) to form GDP.
- In the inactive state, the α subunit binds to GDP and the three subunits are attached together. When the α subunit binds to GTP, its affinity to the $\beta\gamma$ subunits is decreased, resulting in their dissociation
- When a G protein attached to GTP, it is **active**, or “on,” while a G protein that’s bound to GDP is **inactive**, or “off.”



Cycling of G protein between active and inactive states

- In the inactive state, $G\alpha$ binds to GDP (guanosine diphosphate).
- Interaction between $G\alpha$ and the agonist-stimulated receptor causes the release of GDP. GTP then binds to the empty site because its concentration in the cell is higher than GDP.
- The GTP-bound $G\alpha$ has low affinity to $\beta\gamma$ subunits, resulting in their dissociation.
- GTP is hydrolyzed to GDP because $G\alpha$ has GTPase activity.

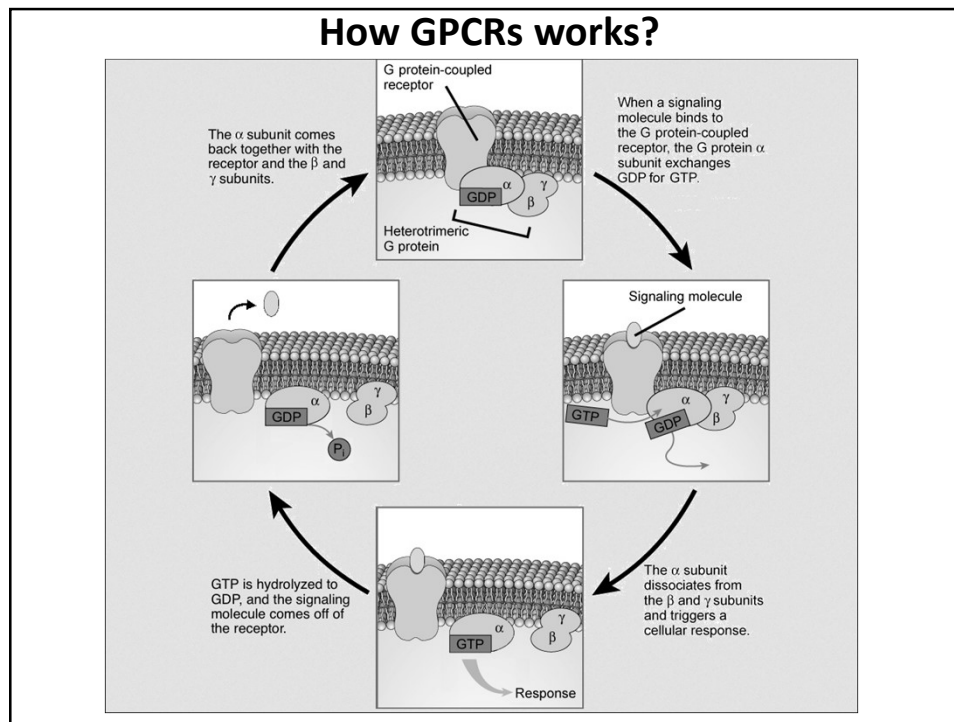
G Protein: Effectors and Second Messengers

• Effectors

- **Effectors** are the target molecules of G protein α or $\beta\gamma$ subunit.
- Table below lists the major effectors of the α subunit. The $\beta\gamma$ subunits may act on adenylyl cyclase, phospholipase A_2 , phospholipase C, ion channels, calcium ATPase etc.
 - **Adenylyl cyclase**
 - Adenylyl cyclase (also known as "adenylate cyclase") catalyzes the conversion of ATP to cAMP, which is an important "second messenger".
 - **Phospholipase A_2**
 - Phospholipase A_2 (PLA_2) cleaves phospholipids, generating arachidonic acid directly.
 - **Phospholipase C**
 - Phospholipase C (PLC) also cleaves phospholipids, but at a different site than PLA_2 , thereby generating diacylglycerol (**DAG**). There are many types of PLC: β , γ , δ and others. The effector of G protein α and $\beta\gamma$ subunits is PLC- β .

• Second messengers

- **Second messengers** are the signaling molecules generated by the stimulation of cell-surface receptors. For example, cAMP, arachidonic acid, DAG and IP_3 generated by the activation of G-protein-coupled receptors are second messengers. The agonists which activate G-protein-coupled receptors are first messengers.



How GPCRs works?

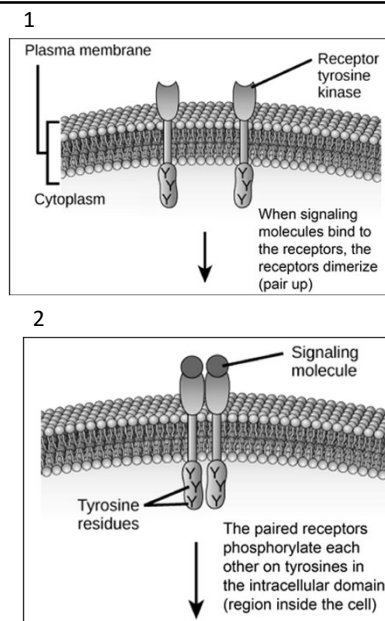
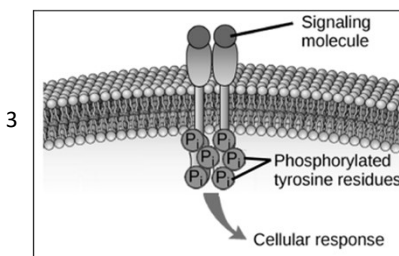
- When its ligand is not present, a GPCR waits at the plasma membrane in an inactive state.
 - G proteins come in different types, but they all bind the nucleotide guanosine triphosphate (GTP), which they can break down (hydrolyze) to form GDP.
 - A G protein attached to GTP is active, or "on," while a G protein that's bound to GDP is inactive, or "off."
- The **G proteins** that associate with GPCRs are a type made up of three subunits, known as **hetero-trimeric G proteins**, namely alpha(α), beta(β) and gamma(γ) subunits.
- When they're attached to an inactive receptor, they're in the "off" form (bound to GDP).
- Ligand binding, however, changes the picture: the GPCR is activated and causes the G protein to exchange GDP for GTP.
- The now-active G protein separates into two pieces (one called the α subunit, the other consisting of the β and γ subunits), which are freed from the GPCR.
- The subunits can interact with other proteins, triggering a signaling pathway that leads to a response.
- Eventually, the α subunit will hydrolyze GTP back to GDP, at which point the G protein becomes inactive. The inactive G protein reassembles as a three-piece unit associated with a GPCR.
- Cell signaling using G protein-coupled receptors is a cycle, one that can repeat over and over in response to ligand binding.

3. Receptor tyrosine kinases (RTK)

- **Receptor tyrosine kinases (RTKs)** are a class of **enzyme-linked** receptors found in humans and many other species.
- They are having an intracellular domains that are associated with an enzyme.
 - In some cases, the intracellular domain of the receptor is an enzyme that can catalyze a reaction.
 - Other enzyme-linked receptors have an intracellular domain that interacts with an enzyme
- A **kinase** is just a name for an enzyme that *transfers phosphate groups to a protein* or other target, and a receptor tyrosine kinase transfers phosphate groups specifically to the amino acid tyrosine.
- In many cases, the phosphorylated receptors serve as a docking platform for other proteins that contain special types of binding domains.
- A variety of proteins contain these domains, and when one of these proteins binds, it can initiate a downstream signaling cascade that leads to a cellular response.

How does RTK signaling work?

- Signaling molecules first bind to the extracellular domains of two nearby receptor tyrosine kinases.
- The two neighboring receptors then come together, or dimerize.
- The receptors then attach phosphates to tyrosines in each others' intracellular domains.
- The phosphorylated tyrosine can transmit the signal to other molecules in the cell.



Significance of Receptor tyrosine kinases (RTKs)

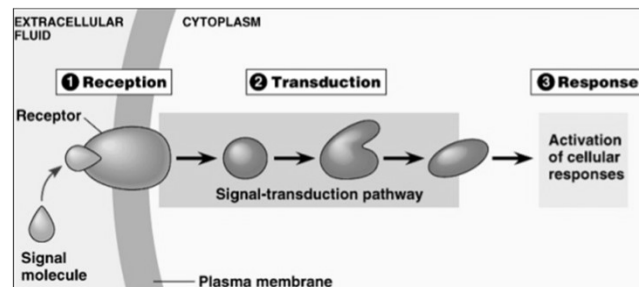
- Receptor tyrosine kinases are crucial to many signaling processes in humans.
 - For instance, they bind to **growth factors**, signaling molecules that **promote cell division and survival**.
 - Growth factors include platelet-derived growth factor (PDGF), which participates in **wound healing**, and nerve growth factor (NGF), which must be continually supplied to certain types of neurons to keep them alive.
- Because of their role in growth factor signaling, receptor tyrosine kinases are essential in the body, but their activity must be kept in balance:
 - overactive growth factor receptors are associated with some types of **cancers**.

Molecules of Signal transduction

- Signal transduction through membrane requires four molecules:
 - **Extracellular Signal Molecule**
 - Produced by one cell and is capable of travelling to receptor site on other cell.
 - **Receptor Protein**
 - Cells must have receptors to receive and bind with signal molecule and communicate its presence inside the cell for triggering the action.
 - **Intracellular Signaling Proteins**
 - They distribute the signals to appropriate parts of the cell. The binding of signal to the receptor activate the intracellular signaling proteins that initiate a signaling cascade.
 - **Target Proteins**
 - The conformations or other properties of the target proteins are altered when a signal transduction pathway is active and changes the behaviour of the responding cell.

Stages of Cell Signaling

1. **Reception:** A cell detects a signaling molecule from the outside of the cell. A signal is detected when the chemical signal (ligand) binds to a receptor protein on the surface of the cell or inside the cell.
2. **Transduction:** When the signaling molecule binds the receptor it changes the receptor protein in some way. This change initiates the process of transduction. Signal transduction is usually a pathway of several steps. Each relay molecule in the signal transduction pathway changes the next molecule in the pathway.
3. **Response:** Finally, the signal triggers a specific cellular response.



THANK YOU